

Neuroscience meets Machine Learning: Encoding Human Brain Activity Through Sparse Autoencoders

Andrea Bragante, Ananda Ferreira Gomes, Yesul Lee

anbra@itu.dk, anaf@itu.dk, yele@itu.dk

Introduction

The human brain is an essential part of our identity, with each brain being unique despite sharing common structures and functions. The specific ways our neurons communicate make us different from one another. Our brains are influenced by genetics, interactions with our surroundings, and both physical and emotional experiences, which strengthen or weaken existing connections and forge new ones. Understanding these unique neural connections can help us comprehend the relationship between the brain, our behaviors, and individual differences. This knowledge is valuable in unraveling the mysteries of human brains and potentially contributing to further neuroscientific research. Significant progress has been made in understanding how the brain processes information over the past 40 years. However, many unknowns remain on individual brain activities (Barry C 2020).

One of the research studied by Kazuo Nishimura presented the idea of how individuals think, act, and possess unique talents, which shows how our brains are activated differently. He emphasized that while each individual is unique in the way they process their thoughts, we can still communicate and interact effectively. To illustrate how we process thoughts differently, his research at Kobe University explored the differences in brain function between visualizers and verbal thinkers. Their research identified that individuals who were strong visualizers exhibited strong activations in these visual areas. On the other hand, verbal thinkers demonstrated weaker activations in these regions, indicating a lesser capacity for visualization. This finding highlights how brain function reflects individual differences in thought processes. However, research aiming to understand these variations — particularly in how thoughts are formed and processed — remains limited. (Nishimura 2021)

We believe that the rise of machine learning and AI techniques can contribute to advancing and enhancing neuroscientific research. By leveraging connections between numerous neurons and their interactions, we can create AI models capable of testing multiple possibilities—many of which are beyond what is comprehensible through traditional physical calculations or current measurement methods.

This new technology can further improve the interpretability of brain functions by identifying patterns through trained models — patterns that couldn't have been recognized in the past without the aid of advanced computational techniques and machine learning algorithms. In our background review, we identified several studies that used AI techniques to explore brain activity, such as predicting brain signals, investigating interactions between brain regions, and reconstructing natural images from Functional Magnetic Resonance Imaging (fMRI) data. However, in our limited exploration of the literature, we found that research using Autoencoders (AE) or Sparse Autoencoders (SAE) to understand brain activity is still in its early stages. We are therefore eager to contribute to the field.

Approach

Although people interpret the world individually, neuroscientific efforts have proven that the brain consists of regions specialised in processing certain types of stimuli. As an example, the visual cortex interprets images, and its subregions recognise visual patterns such as written words, faces, bodies and other properties (Huff T 2023). The goal of our study is to explore whether an unsupervised model, such as an Autoencoder, could similarly recognize patterns in brain activity and meaningfully encode them.

For our project, we selected an extensive collection of fMRI data, consisting of Blood Oxygenation Level Dependent (BOLD) signals. We focus on the brain region specialised in visually processing human bodies and limbs. To understand brain activity, we train a Sparse Autoencoder to extract features that capture brain response activated by looking at different images. Specifically, we focus on the differences between images depicting people and scenes without people.

We believe that creating a successful model capable of recognising visual patterns enables a more meaningful understanding of brain activity, along with the possibility to perform neuroscientific experiments without directly involving the physical brain.

This is why we consider our project's domain an important topic to explore: it could serve as a small step forward in advancing research at the intersection of Machine Learning and neuroscience.

Dataset

The dataset is a subset of the Natural Scenes Dataset (NSD) as provided by the Algonauts Challenge 2023 (AC). NSD is a massive high-resolution fMRI dataset, where the brain responses of 8 subjects were scanned while exposed to visual stimulation in the form of natural scene images. The data was collected in 30-40 sessions per subject within a year, throughout which each participant was shown 9000-10000 distinct color images. 1000 of those images were shared among all subjects, while the remaining were exclusive to each subject. For simplification purposes, we will refer to them as *shared* and *specific* images from here on.

All images used in NSD are retrieved from the Common Objects in Context (COCO) dataset, which holds the relevant metadata to identify images by category. In our case, we are interested in pictures that depict human bodies and are therefore categorized as ‘person’, the reasoning is further explained in the following subsection. A detailed description of the acquisition and pre-processing of the dataset is available in the NSD Paper (Allen et al. 2022).

AC data consists of a subset of the fMRI data, originally available for the full volume of the brain, that is limited to the cortical surface vertices of the visual cortex, which is the area of the brain responsible for interpreting visual stimulation (Huff T 2023). Since this research is focused on processing visual information, this narrowed-down version of the dataset is sufficient for our purpose. The left and right hemisphere (LH and RH) data are given separately, with 19.004 and 20.544 cortical surface vertices respectively, except for subjects 6 (18.978 LH and 20.220 RH vertices) and 8 (18.981 LH and 20.530 RH vertices) due to missing data. This allows us to focus on the LH, simplifying the data for training and analysis. The fMRI data is further projected to FreeSurfer’s fsaverage surface, an average representation of the human cortical surface with 163.842 vertices per hemisphere (Developers 2024b).

Regions of Interest and Classes

ROIs are areas in the brain that are defined based on their functional properties. Each person has a unique brain structure. This is why the location, the spatial relation, and the count of the corresponding voxels of ROIs vary depending on the participant. Along with the fMRI data and the corresponding stimuli, AC includes ROI indices and mapping of those indices to the individual surface vertices of each participant and hemisphere. Each ROI is assigned to one of the following classes: *prf-visualrois*, *floc-bodies*, *floc-faces*, *floc-places*, *floc-words*, and *streams*. For a more targeted analysis, we chose to focus on the class *floc-bodies* and simplify the data to only contain the corresponding surface vertices. The *floc-bodies* region is composed of the four ROIs: *EBA*, *FBA-1*, *FBA-2*, and *mTL-bodies*, which are specialised in processing images of human bodies and limbs (Allen et al. 2022).

The vertices belonging to the *floc-bodies* class were identified for each participant individually through the NSD functional localizer (fLoc) experiment (Allen et al. 2022). Not all subjects have all ROIs in each hemisphere. As an example, *mTL-bodies* is only present in the sixth and eighth



Figure 1: Posterior-ventral view of the floc-bodies ROIs of each subject, LH and RH; Red: EBA; Green: FBA-1, Blue: FBA-2, Purple: mTL-bodies

subject’s right hemisphere, while the second and seventh subjects have only *EBA* and *FBA-2* in both hemispheres. It is notable that the ROI *EBA*, which is responsible for visually identifying human bodies (Downing et al. 2006), is dominantly present in all subjects and hemispheres. Further differences can be seen in Figure 1.

With the assumption that stimuli categorised as *person* trigger higher activations within the *floc-bodies* region than natural scene images without people, we decided to conduct our analysis with fMRI responses recorded for *person* images in comparison to those recorded for images without humans, namely *non-person* images. To investigate this assumption, we visualised the *floc-bodies* fMRI responses of the first five subjects when exposed to the same image. For all participants, the neural activations are the highest for the image with multiple people in focus and minimal for images without any humans (Figure 2). This supports our expectations, making the *person* category and *floc-bodies* class a suitable pair of hyperparameters for our analysis.

Train, Validation, and Analysis Split

To construct a model of each participant’s brain, we use their fMRI signals as the input for our SAE. For each subject, AC provides fMRI data only within the challenge’s training split of the dataset, the test split is withheld from the public. We therefore use the available training split for training, validation, and analysis in our research, resulting in a total of [9841, 9841, 9082, 8779, 9841, 9082, 9841, 8779] fMRI samples per subject. All LH fMRI vectors are masked to the *floc-bodies* class, limiting the surface vertices to [3411, 4117, 3633, 4325, 5114, 4607, 4295, 4056] for the eight subjects.

Per subject, the fMRI data is split into samples retrieved from stimulation by *specific* and *shared* images. The former is used for training and validation with a distribution of 90% and 10%. The latter is reserved for our analysis to

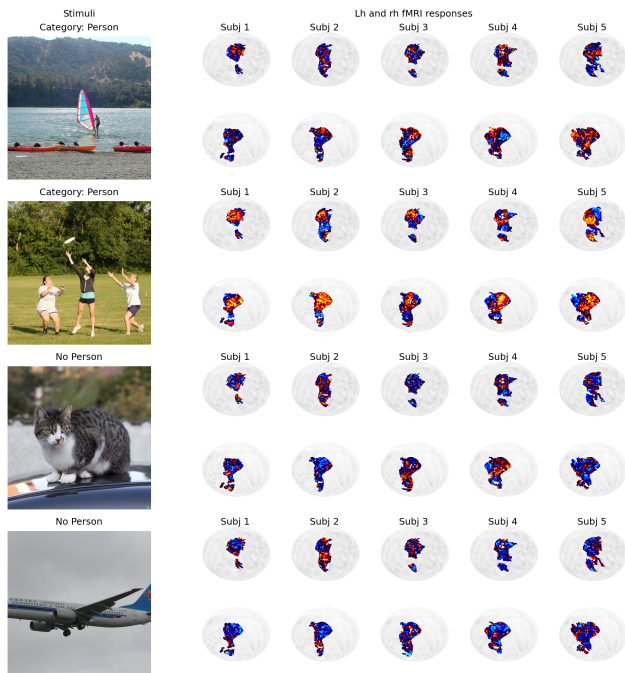


Figure 2: LH (top) and RH (bottom) floc-bodies activations of subject 1 to 5 for four samples of shared images.

ensure common ground when comparing subjects and the latent representation of their brain activity. The analysis split makes up around 10% of the full dataset. It is again divided into *person* and *non-person* samples, whereas both subsets are sliced to the same length of 382 samples, the minimum amount available among all subjects. The train and validation split show an almost even distribution between *person* and *non-person* data for all subjects. This ensures that all models are equally trained to extract features for both categories, a desirable precondition for the subsequent analysis.

Data distribution

The fMRI data are BOLD signals and range from positive to negative values. Depending on subjects, the amplitude of values differs. The distribution, however, is almost equal between subjects, with a median around 0.03, the middle 50% of the data lying between -0.7 and 0.6, and adjacent values being around -2.6 and 2.6. The approximated values are true for both the train and validation split. By this we can assume that the data distribution won't negatively impact the autoencoder's performance when training for different subjects.

Background

Since the publishing of NSD, several studies have been done, demonstrating how large-scale, high-quality fMRI data can drive advancements in the intersection of neuroscience and machine learning.

To test the strength of NSD, (Allen et al. 2022) trained a convolutional network based on the encoding method by (Naselaris et al. 2011). The study introduced 'G-Net', a

brain-optimized CNN trained using stimuli images as input to predict fMRI responses. For comparison, the researchers employed a pre-trained AlexNet, which is a task-optimized CNN developed by (Krizhevsky, Sutskever, and Hinton 2012). Given the high quality of the NSD data, the brain-optimized G-Net outperformed the task-optimized AlexNet. However, with a smaller set of training data, G-Net showed poor performance. This performance gap highlighted the importance of the NSD dataset, which played a pivotal role in this research by providing the necessary scale and quality of data.

Also we found other articles exploring Graph Neural Networks (GNNs), which seem to capture both structural information and functional connectivity in the brain (Wein et al. 2021). Research has shown the capabilities of GNNs for representation learning, while Graph Convolutional Network-derived architectures, namely Graph Autoencoders (GAE), have been proposed and studied in the context of fMRI data usage (Noman et al. 2024).

In (Naselaris et al. 2009), BOLD fMRI measurements from early and anterior visual areas were incorporated into a Bayesian decoder to reconstruct complex natural images. We found this paper relevant to our research, as it highlights how technological advancements can enhance the measurement of brain signals and enable brain-technology integration. Such progress has the potential to improve various fields related to neuroscience.

There is little research on the effects of sparsity for getting an optimized set of features to study neural processes, including (Mahmoud, Karamti, and Alrowais 2020) and (Shu and Fyshe 2013). However, there have been advancements in the study of interpretability in other domains, especially for Large Language Models (LLM). In a recent study, (Cunningham et al. 2023) use Sparse Autoencoders to reduce superposition in LLMs' neural activations, enhancing interpretability.

Methods

An Autoencoder is a model usually trained to reconstruct its input from an encoded representation. It compresses the input's most relevant features into a lower dimensional latent vector, which is then decoded targeting its original state. The dimensionality reduction, however, poses the disadvantage of superposition, where a limited number of neurons represent a higher amount of features. Each neuron activates for multiple semantically unrelated features, making the neural network difficult to understand (Cunningham et al. 2023). This can be relieved by allowing higher dimensionality and introducing sparsity to the network. With an L1 regularisation, sparsity can be enforced onto the latent space to guide the model to activate neurons in a more interpretable way (Gao et al. 2024). We can further introduce sparsity to the model parameters, which reduces computational overhead and strengthens the learning (Enthusiast 2024).

Exploring the architecture

As a starting point, we implement a vanilla autoencoder with a simple Multi Layer Perceptron (MLP) architecture, adding

layers of complexity as we proceed. We select one test subject and experiment with different numbers of hidden layers and their dimensionalities, using Mean Squared Error (MSE) for our loss function. Running some quick tests on a few epochs, we find reasonable loss results using an encoder with two dense layers, the sizes of which are a third and a sixth of the input data size, respectively. This way, the input can be proportionally reduced in dimensionality even when training on different data. Considering that the input size varies depending on each subject amount of ROI class vertices, this is a practical decision for building a scalable model. Applying this ratio gives good results for the test subject, by which we can assume the model would perform similarly for all subjects. The model decoder mirrors the architecture of the encoder, as typically done with autoencoders. ReLU is used after each dense layer for non-linearity.

The fMRI data is complex and difficult to interpret, which makes having a low reconstruction loss a difficult task for the autoencoder. We decide to test the model’s learning ability on two smaller datasets, MNIST (introduced by (Lecun et al. 1998)) and CIFAR10 (introduced by (Krizhevsky 2009)). Successful results reassure us that we built a functioning architecture, allowing us to rely less on the loss metrics, and instead inspect the input reconstruction manually by mapping the output vector back into the fsaverage surface space. The visualisations of the reconstructed fMRI signals show activations in the regions that were also activated in the original data (Figure 3). Since the scope of our analysis is ROI- and not voxel-based, we are satisfied with the extent to which the model decodes the fMRI. Our goal is not to train a model that reproduces its input perfectly, as is often the case when using autoencoders, but rather to train one that encodes the input into a more interpretable latent representation. After all, an autoencoder with a minimal reconstruction error does not necessarily succeed in extracting the most relevant features for the task (Braun et al. 2024).

Sparsity

After establishing a working vanilla autoencoder, we implement sparsity, both in the parameters and latent space. In the parameters, constant sparsity is enforced through a hyperparameter using Jaxpruner, a library developed by the Google Jax team that allows for a simple implementation in a Jax workflow (Lee et al. 2023). To penalise the activations in the latent representation, we introduce an L1 regularisation on the latent vector z of dimension M to the loss function:

$$\begin{aligned} \text{total}_{\text{loss}} &= \text{MSE}_{\text{loss}} + \text{sparsity}_{\text{loss}} \\ &= \frac{1}{N} \sum_{i=1}^N (\hat{x}_i - x_i)^2 + \lambda \cdot \frac{1}{M} \sum_{j=1}^M |z_j|. \end{aligned}$$

The impact of sparsity on the loss function is controlled by a hyperparameter, the L1 coefficient λ . MSE acts as the reconstruction loss, where x is the input and $\hat{x}$ its reconstruction for N numbers of input samples. Furthermore, the ReLU activation function is used to zero-out negative values in the last layer of the encoder.

At this point, we have a set of relevant hyperparameters to work with and try out various combinations of values to

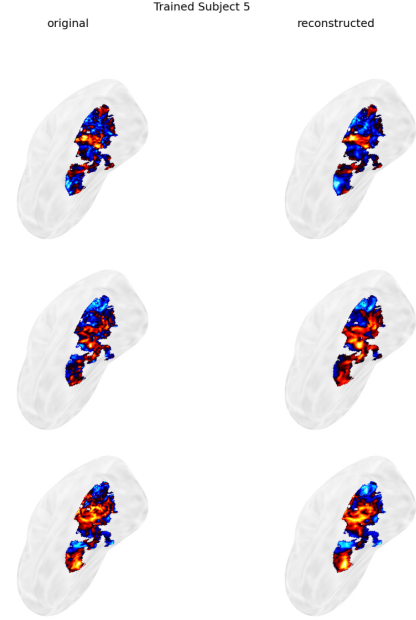


Figure 3: Posterior-ventral view of original and reconstructed LH floc-bodies activations of subject 5 for 3 samples. Left: original floc-bodies activations as acquired by NSD; right: reconstructed floc-bodies activations as output by the model trained with low sparsity and latent size 32.

find out what works best. We run different experiments with several parameter combinations:

- Learning rate of 0.0001, 0.0003, and 0.001
- Sparsity constant of 0, 0.8, and 0.85
- L1 coefficient of 0.1 and 0.01
- Latent dimensions of 16, 32, and 64

The experiments show that higher sparsity typically results in a less successful loss minimisation. However, as stated before, reconstruction is not our model’s main task and we decide to compromise on loss reduction if it means that we could retrieve more meaningful feature activations.

Final selection of experiments

For the above reason, we decide to perform the actual production training with the following set of parameters: a batch size of 256, a learning rate of 0.0001, and training for 150 epochs. Since we want to explore the impact of latent dimension and sparsity, we decide to train four models for each subject, using these combinations:

- Small latent size (32), low sparsity (L1 coefficient=0.01, sparsity=0.8)
- Small latent size (32), high sparsity (L1 coefficient=0.1, sparsity=0.85)
- Large latent size (64), low sparsity (L1 coefficient=0.01, sparsity=0.8)
- Large latent size (64), high sparsity (L1 coefficient=0.1, sparsity=0.85)

	Low sparsity		High sparsity	
	32	64	32	64
Latent size	32	64	32	64
Avg. duration	5m 6s	6m 53s	4m 56s	4m 50s
Avg. train loss	0.327	0.27	0.4	0.359
Avg. eval loss	0.347	0.303	0.447	0.378

Table 1: Training statistics for our final experiments

We logged the training process with AimStack (Developers 2024a).

Analysis methods

After the training process, we load the fMRI data of the shared images and use our model instances to encode the brain activity of each participant. For each subject, we obtain two sets of latent vectors, one for each category, *person* and *non-person*. Each set contains 382 vectors of size 32 or 64, depending on the experiment.

The following steps are all done for each subject and experiment. First, we check whether there is a significant difference between the two distributions, *person* and *non-person*. We check with the *Multivariate Normality Test* if they follow a normal distribution, in order to decide how to test for significance. Since the data turns out to be not normally distributed, we run a Mann–Whitney U test on the two sets, since it does not assume normality in the data distribution.

We finally use t-Distributed Stochastic Neighbor Embedding (t-SNE) to visualise the distribution of encoded *person* and *non-person* features. This method preserves the relationships between neighboring data points by finding similarities and placing them closely in the lower-dimensional space (Soni 2024). The expectation is that the SAE would extract interpretable features from the input fMRI signals, that represent information about identifying a person or not. Therefore, it is crucial to maintain the relation between the sparsely activated neurons of the latent vectors when trying to put those activations into clusters.

Results

The models work well, although it can be difficult to understand how well. The performance of our models is constrained not only by the restrictive choice of MLP as underlying base, but also by the time and computational power at our disposal. For the following results, we refer to the four specific experiments we introduced in the section Methods, which are the ones we analyse.

The training process for each individual autoencoder takes, on average, about 5 minutes and 20 seconds on a MacBook Pro M3, reaching a total time of 2 hours and 54 minutes. Table 1 contains the average duration, training loss, and validation loss for the four experiments.

Figure 4 shows the training loss curves for the four experiments and makes clear that, as sparsity constraints decrease and latent dimension increases, the loss gets smaller: it is easier for the model to learn without a sparsity constraint and being able to use a larger latent space.

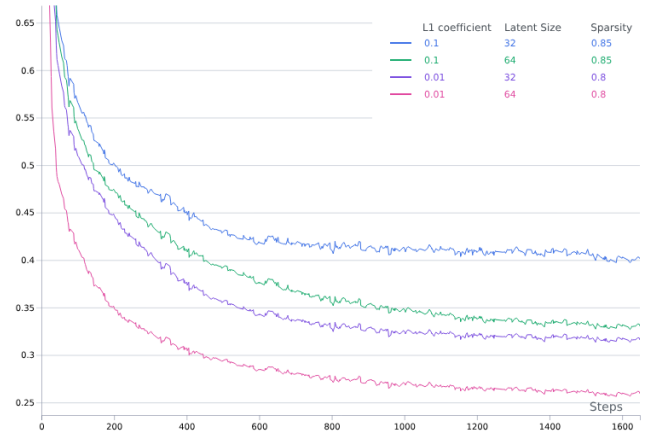


Figure 4: Training curves for our final experiments

	Low sparsity		High Sparsity	
	32	64	32	64
1	1.23⁻⁶	0.0001	0.005	8.08⁻¹⁷
2	0.32	6.27⁻⁵	0.003	7.01⁻¹¹
3	0.79	0.06	1.34⁻⁵	0.01
4	0.16	0.08	0.01	0.24
5	4.15⁻⁵	1.64⁻⁵	1.78⁻⁹	7.87⁻¹⁸
6	0.0003	0.46	0.27	0.76
7	7.21⁻⁸	0.57	1.50⁻⁹	2.36⁻⁵
8	4.87⁻⁶	0.75	0.001	0.07

Table 2: P-values, per experiment, per subject

To visualise the impact of sparsity on our resulting latent spaces, Figure 5 shows latent activations resulting from the inference with the first shared image on the trained model for the first three subjects.

Table 2 reports the p-values resulting from performing a *Mann–Whitney U* test on *person* vs *non-person* distributions, for each subject and for each experiment. The values in bold are the significant ones, i.e. those whose p-values are ≤ 0.05 .

Finally, Figure 6 show the data visualised using t-SNE; in the plots we can see the distributions for *person* and *non-person* data, with an indication on whether the difference in that particular subject was or was not significant.

We also sample the same number of vectors from a normal distribution and plot it in Figure 7 with t-SNE, as a sort of baseline to keep from over-interpreting what we see in our data.

Even on a first glance, there is some clear difference between our data and the normal data plot. The randomly sampled data has a pretty well-defined circular shape, and shows a dense, pretty uniform amount of dots.

Our plot, on the other hand, shows more irregular shapes, and a less uniform spread of dots across the two distributions. There is a notable variance between different subjects. In particular, all subjects show a clearer separation between the two colors, and some more than others, for example sub-

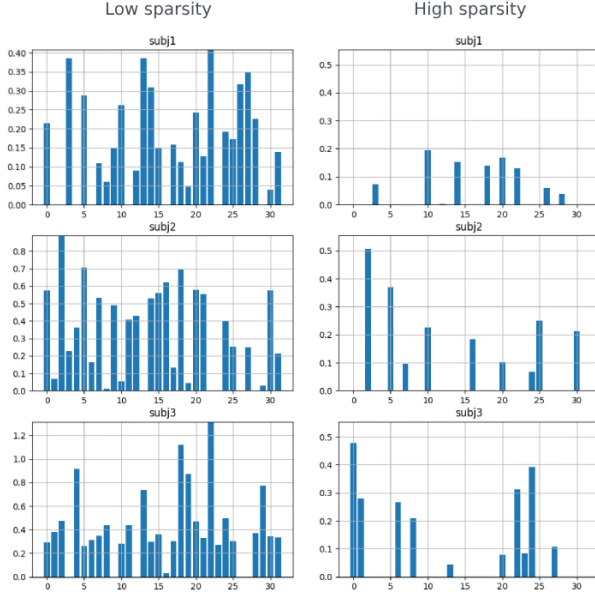


Figure 5: Latent activations with low and high sparsity

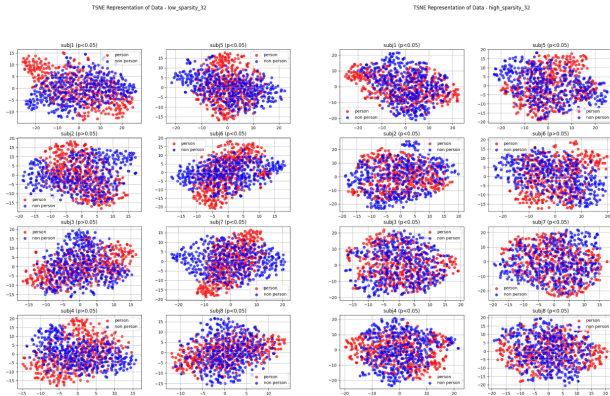


Figure 6: t-SNE visualisations for low and high sparsity, latent size of 32

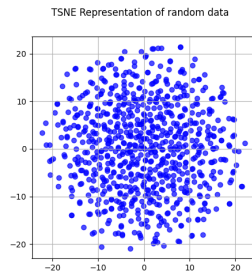


Figure 7: t-SNE visualisation of a Normal distribution

jects 5 and 8: some areas in the plot contain a larger number of dots of one color and none of the other, which appears unlikely to be random. While we can see a pattern in how distributions separate, we cannot directly compare the subjects. For each subject, we train a different instance of our autoencoder model, and we do nothing to enforce all the latent spaces to be comparable.

We want to be cautious with our findings, as we might be biased towards over-interpreting what we see as patterns, and this research is in its early stages. The category separation is fairly modest, and if we were to train a classifier to predict whether a sample would belong to one or another category, the accuracy would probably be fairly low. That being said, there is a clear separation between the two distributions in each of our plots, hinting to the distinction between the two categories, and suggesting that our model can separate between them.

Discussion

The biggest strength of our approach is the ease of implementation and versatility. Looking at the bigger picture, our model is a vanilla autoencoder using MLP for its components, which could be almost effortlessly reimplemented with different libraries and use different datasets, as in fact we did with MNIST and CIFAR10.

This provides us with a tweakable baseline, which we can then use with elaborate data such as fMRI. In the future, we could start from here to explore more sophisticated methods and approaches.

The structures we use for preprocessing, for example to read the dataset or correlate fMRI activations with COCO categories, allow us to experiment with the scope and brain regions to focus on. This modularity is a feature that enhances scalability in the analysis. While in this project we decided to focus on the *floc-bodies* class in the LH, at some point in the future it would be desirable to confirm findings for one ROI class by investigating other regions, as well as test interpretability within a bigger scope.

Exploring other architectures

This ease of implementation is a double-face medal. Because of its simplicity, it faces a bottleneck in its very whole fundamental idea, that is the use of a simple MLP as base for the AE architecture. Models used for image and object recognition tasks seem to perform better when making use of convolutional layers. Convolutional operations are a highly efficient way to capture spatial features and patterns (LeCun, Bengio, and Hinton 2015), and hence we are positive that our model would benefit from that. FMRI scans are typically stored as a four-dimensional NIfTI image, where the spatial relation between the brain voxels are maintained. We faced the issue that the fMRI data provided by the Algonauts Challenge is flattened into a one-dimensional vector of voxels. Despite having the means to map this vector onto the fsaverage surface, we had trouble converting it to a 2D or 3D structure that we could run convolutions on. Our next step would be to expand our model by implementing CNNs.

A method we noticed in literature but did not study in

detail is GNNs and GAEs. Since the brain is a complex network of interconnected regions an effective way to represent its structure preserving both its topology and functional connections, is to use a graph. As with convolutions, it would be a natural next step to expand our research by exploring GNNs as part of the AE architecture. A downside specific to our project is that the full power of these architectures comes into play when dealing with the whole brain, given the more extended graph structure; since we decided to focus on only one ROI class, it might be excessive to use GAE on such a small set of data.

Future work

Our future work agenda would be to expand dramatically the scale of experimentation for what concerns model architectures, hyperparameter values, visualisation methods, and statistical analyses methods.

This project was a small introductory approach to the use of autoencoders, and in particular sparse autoencoders, to meaningfully encode complex fMRI data into a more interpretable structure. We barely scratched the surface of interpretability of neural networks: the way we used sparsity allows us to have latent vectors potentially more interpretable, but as far as we implemented it now, sparsity is more of a regularisation method. It has a positive impact on our training process, but we didn't come to the point where we could accurately test its benefits and exploit it for interpretability.

With a larger time window, we would explore ways to enforce the latent space of all participants to be comparable, and then investigate if some interesting patterns emerge across them in the way the model interprets the fMRI data.

Conclusion

All being said, we are positive that our project looks promising. Our model was not explicitly trained on separating between the two categories, and it encodes information about highly complex fMRI data into a small vector. Yet, performing inference on known-category data, we observe our model to output vectors fairly differently based on the category.

This does not allow us to state that it specifically encodes information about this categorization. Natural scene images present a large spectrum of different features, which in turn result in complex activation patterns in the brain. Those features most likely overlap for person and non-person stimuli, making it difficult for the model to distinguish categories.

Yet, this is a very small scale experiment, and we specifically selected floc-bodies because it is known in the literature to be sensitive to stimuli containing bodies. Our conclusion is, therefore, that our results suggest that our sparse autoencoder, trained on activations within the floc-bodies region, might be able to capture information about whether the participant was looking at an image with or without a human body in it.

References

- Allen, E. J.; St-Yves, G.; Wu, Y.; Breedlove, J. L.; Prince, J. S.; Dowdle, L. T.; Nau, M.; Caron, B.; Pestilli, F.; Charest, I.; Hutchinson, J. B.; Naselaris, T.; and Kay, K. 2022. A massive 7T fMRI dataset to bridge cognitive neuroscience and artificial intelligence. *Nat. Neurosci.* 25(1):116–126.
- Barry C, McMillan F, W. A. 2020. why study the brain. *The Brain* 5. Accessed: 2024/12/10.
- Braun, D.; Taylor, J.; Goldowsky-Dill, N.; and Sharkey, L. 2024. Identifying functionally important features with end-to-end sparse dictionary learning.
- Cunningham, H.; Ewart, A.; Riggs, L.; Huben, R.; and Sharkey, L. 2023. Sparse autoencoders find highly interpretable features in language models.
- Developers, A. S. 2024a. Aim: An open-source ai metadata tracker. <https://aimstack.io/>. Accessed: 2024-12-12.
- Developers, M.-P. 2024b. Background on freesurfer.
- Downing, P. E.; Peelen, M. V.; Wiggett, A. J.; and Tew, B. D. 2006. The role of the extrastriate body area in action perception. *Social Neuroscience* 1(1):52–62.
- Enthusiast, E. H. 2024. A review of sparse expert models in deep learning. <https://medium.com/@EleventhHourEnthusiast/a-review-of-sparse-expert-models-in-deep-learning-5b29611da9d9>. Accessed: 2024-12-12.
- Gao, L.; la Tour, T. D.; Tillman, H.; Goh, G.; Troll, R.; Radford, A.; Sutskever, I.; Leike, J.; and Wu, J. 2024. Scaling and evaluating sparse autoencoders.
- Huff T, Mahabadi N, T. P. 2023. Neuroanatomy, visual cortex. *Neuroanatomy, Visual Cortex*. Accessed: 2024/12/12.
- Krizhevsky, A.; Sutskever, I.; and Hinton, G. E. 2012. Imagenet classification with deep convolutional neural networks. *Communications of the ACM* 60(6):84–90.
- Krizhevsky, A. 2009. Learning multiple layers of features from tiny images. 32–33.
- LeCun, Y.; Bengio, Y.; and Hinton, G. 2015. Deep learning. *Nature* 521(7553):436–444.
- Lecun, Y.; Bottou, L.; Bengio, Y.; and Haffner, P. 1998. Gradient-based learning applied to document recognition. *Proceedings of the IEEE* 86(11):2278–2324.
- Lee, J. H.; Park, W.; Mitchell, N.; Pilault, J.; Obando-Ceron, J.; Kim, H.-B.; Lee, N.; Frantar, E.; Long, Y.; Yazdanbakhsh, A.; Agrawal, S.; Subramanian, S.; Wang, X.; Kao, S.-C.; Zhang, X.; Gale, T.; Bik, A.; Han, W.; Ferev, M.; Han, Z.; Kim, H.-S.; Dauphin, Y.; Dziugaite, G. K.; Castro, P. S.; and Evci, U. 2023. Jaxpruner: A concise library for sparsity research.
- Mahmoud, A.; Karamti, H.; and Alrowais, F. 2020. An effective sparse autoencoders based deep learning framework for fMRI scans classification. In *Proceedings of the 22nd International Conference on Enterprise Information Systems*. SCITEPRESS - Science and Technology Publications.
- Naselaris, T.; Prenger, R. J.; Kay, K. N.; Oliver, M.; and Gallant, J. L. 2009. Bayesian reconstruction of natural images from human brain activity. *Neuron* 63(6):902–915.
- Naselaris, T.; Kay, K. N.; Nishimoto, S.; and Gallant, J. L. 2011. Encoding and decoding in fmri. *NeuroImage* 56(2):400–410.

Nishimura, K. 2021. The way we think is reflected in our brain activity. *Research Outreach* 121. Accessed: 2024/12/10.

Noman, F.; Ting, C.-M.; Kang, H.; Phan, R. C.-W.; and Ombao, H. 2024. Graph autoencoders for embedding learning in brain networks and major depressive disorder identification. *IEEE Journal of Biomedical and Health Informatics* 28(3):1644–1655.

Shu, M., and Fyshe, A. 2013. Sparse autoencoders for word decoding from magnetoencephalography.

Soni, S. 2024. Mastering t-sne: t-distributed stochastic neighbor embedding. <https://medium.com/@sachinsoni600517/mastering-t-sne-t-distributed-stochastic-neighbor-embedding-0e365ee898ea>. Accessed: 2024-12-12.

Wein, S.; Malloni, W. M.; Tomé, A. M.; Frank, S. M.; Henze, G. I.; Wüst, S.; Greenlee, M. W.; and Lang, E. W. 2021. A graph neural network framework for causal inference in brain networks. *Scientific Reports* 11(1).